Mathematical and Computational Methods

Lecture 4: Vectors

With this lecture the module changes subject. The first three lectures built the foundations—numbers, functions and the complex plane; the next three open the *linear-algebra* strand, whose objects are not single numbers but *vectors* and the arrays—matrices—that act on them, with the calculus following in the second half of the course. A vector carries both a size and a direction, as many quantities in geometry, mechanics and data do: a displacement, a velocity, a force, or a list of measurements. This lecture introduces vectors in the plane and in space, their algebra (addition, scaling, magnitude), the two ways of *multiplying* them—the scalar (dot) and vector (cross) products—and the equations of *lines* and *planes* that they let us write down.

Where this fits. The coordinate plane and elementary geometry of Lecture 1 are the only prerequisites; no calculus is needed, so the linear-algebra units stand on their own. The dot product defined here generalises in one line to $\mathbb{R}^n$ and supplies the *cosine similarity* the data-science thread has been pointing towards. Two ideas are left at the level of *intuition* and made computational in Lecture 5: testing vectors for independence, and—hiding inside the cross product—the *determinant*, used here only as a memory device and defined properly next time. The lines and planes of Section 3 are solution sets of linear systems, so their intersections return as the geometry of Lecture 5, and the strand closes with eigensystems in Lecture 6.

Contents

Learning objectives

By the end of this lecture you will be able to:

- Add, subtract and scale vectors, and find magnitudes and unit vectors.
- Compute the dot product (angle, projection, orthogonality) and the cross product (normal, area).
- Derive and use the equations of lines and planes.
- Explain linear independence and basis, and interpret cosine similarity in $\mathbb{R}^n$.

1 Vectors and their algebra

1.1 Scalars and vectors

A *scalar* is a quantity described by a single number—a temperature, a mass, a length. A *vector* needs a number *and* a direction: a wind of 30 km/h *towards the north-east*, or a force of 10 N *downward*. We draw a vector as an arrow whose length is its magnitude and whose heading is its direction.

Definition 1.1 (Vector). Fix coordinate axes. A *vector* is an ordered list of real numbers, its *components*,

$$\mathbf{v} = (v_1, v_2) \in \mathbb{R}^2, \qquad \mathbf{v} = (v_1, v_2, v_3) \in \mathbb{R}^3, \qquad \text{or} \qquad \mathbf{v} = (v_1, \ldots, v_n) \in \mathbb{R}^n.$$

Vectors with the same number of components (the same *dimension*) are added, and a vector is scaled by a real number k, *componentwise*:

$$\mathbf{u} + \mathbf{v} = (u_1 + v_1, \ldots, u_n + v_n), \qquad k\mathbf{v} = (kv_1, \ldots, kv_n).$$

What makes $\mathbb{R}^n$ a space of vectors is exactly that it is *closed* under these two operations: adding two vectors, or scaling one, again gives a vector of the same kind. The *zero vector* $\mathbf{0} = (0, \ldots, 0)$ is the identity for addition.

We write vectors in bold, $\mathbf{u}, \mathbf{v}, \mathbf{w}$, and scalars in ordinary italic, a, k, t. (By hand, an underline $\underline{v}$ or an arrow $\vec{v}$ is usual.)

Note (Geometric reading). In $\mathbb{R}^2$ and $\mathbb{R}^3$ a vector is pictured as a *directed segment* (an arrow) from an *initial point* to a *terminal point*. Two arrows with the same length and direction—one a parallel translate of the other—represent the *same* vector; sliding $\mathbf{v}$ so its tail is at the origin, the coordinates of its tip are the components above.

Note (Intuition, not the definition). It is often helpful to think of a nonzero vector as carrying a *magnitude* and a *direction*; but that picture is only a label for the data above, and it is the two operations—addition and scaling—not the words "magnitude" and "direction"—that make something a vector. (The abstract notion of a *vector space*, of which $\mathbb{R}^n$ is the first example, is developed in the Linear Algebra module.)

1.2 Components

A vector's components are read off the picture exactly as above: slide it so its initial point is the origin, and the coordinates of its terminal point *are* the components. Conversely, any two points determine a vector.

Definition 1.2 (The vector between two points). The vector from a point $P(p_1, p_2, p_3)$ to a point $Q(q_1, q_2, q_3)$ is found by subtracting coordinates,

$$\overrightarrow{PQ} = (q_1 - p_1,\ q_2 - p_2,\ q_3 - p_3),$$

with the obvious two-term version in $\mathbb{R}^2$.

Note (The standard basis). Writing $\mathbf{i} = (1,0,0)$, $\mathbf{j} = (0,1,0)$, $\mathbf{k} = (0,0,1)$ for the unit vectors along the axes, every vector in $\mathbb{R}^3$ is built from them in exactly one way:

$$\mathbf{v} = (v_1, v_2, v_3) = v_1\mathbf{i} + v_2\mathbf{j} + v_3\mathbf{k}.$$

The two notations—triples and $\mathbf{i}, \mathbf{j}, \mathbf{k}$ combinations—are interchangeable; in $\mathbb{R}^2$ one drops the third entry and uses $\mathbf{i}, \mathbf{j}$.

1.3 Addition, subtraction and scalar multiplication

Addition and scalar multiplication were defined componentwise in the definition of a vector; each has a clean geometric meaning, and *subtraction* is the one further operation we need.

Definition 1.3 (Subtraction). For $\mathbf{u} = (u_1, u_2, u_3)$ and $\mathbf{v} = (v_1, v_2, v_3)$, subtraction is addition of the negative,

$$\mathbf{u} - \mathbf{v} = \mathbf{u} + (-1)\mathbf{v} = (u_1 - v_1,\ u_2 - v_2,\ u_3 - v_3).$$

Geometrically, $\mathbf{u} + \mathbf{v}$ is found by the *triangle rule*—place the tail of $\mathbf{v}$ at the tip of $\mathbf{u}$ and join start to finish—equivalently the diagonal of the parallelogram on $\mathbf{u}$ and $\mathbf{v}$ (Figure 1). Scalar multiplication $k\mathbf{v}$ stretches $\mathbf{v}$ by the factor $|k|$, keeping its direction if $k > 0$ and reversing it if $k < 0$.

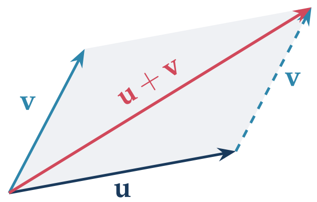

Figure 1: The triangle (and parallelogram) rule for $\mathbf{u} + \mathbf{v}$.

Example 1.1 ***(Component arithmetic).*** With $\mathbf{u} = (1,3)$ and $\mathbf{v} = (4,-1)$,

$$\mathbf{u} + \mathbf{v} = (5,2), \qquad \mathbf{u} - \mathbf{v} = (-3,4), \qquad 2\mathbf{u} = (2,6).$$

These operations obey the rules one expects, proved in each case by checking one component at a time.

Proposition 1.1 (Algebraic properties). For all vectors $\mathbf{u}, \mathbf{v}, \mathbf{w}$ and scalars a, k,

$$\mathbf{u}+\mathbf{v}=\mathbf{v}+\mathbf{u}, \quad (\mathbf{u}+\mathbf{v})+\mathbf{w}=\mathbf{u}+(\mathbf{v}+\mathbf{w}), \quad \mathbf{u}+\mathbf{0}=\mathbf{u}, \quad \mathbf{u}+(-\mathbf{u})=\mathbf{0},$$

$$k(\mathbf{u}+\mathbf{v})=k\mathbf{u}+k\mathbf{v}, \quad (a+k)\mathbf{v}=a\mathbf{v}+k\mathbf{v}, \quad a(k\mathbf{v})=(ak)\mathbf{v}, \quad 1\,\mathbf{v}=\mathbf{v}.$$

1.4 Magnitude and unit vectors

The length of a vector is read straight off its components by Pythagoras (Figure 2).

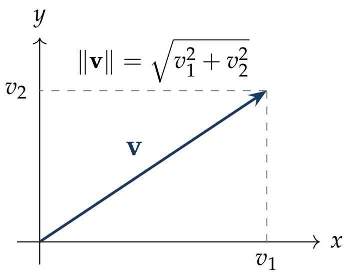

Figure 2: A vector and its components; the magnitude is the hypotenuse.

Definition 1.4 (Magnitude). The *magnitude* (or *norm*, or *length*) of $\mathbf{v}=(v_1, v_2, v_3)$ is

$$\|\mathbf{v}\| = \sqrt{v_1^2+v_2^2+v_3^2},$$

with the obvious two-term version in $\mathbb{R}^2$. The distance between points P and Q is $\left\|\overrightarrow{PQ}\right\|$.

A vector of length one is a *unit vector*; it records a pure direction. Dividing any nonzero vector by its own length produces the unit vector pointing the same way, a step called *normalising*.

Definition 1.5 (Unit vector). For $\mathbf{v} \neq \mathbf{0}$, the unit vector in the direction of $\mathbf{v}$ is

$$\hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|}.$$

Example 1.2 *(Normalising)*. For $\mathbf{v}=(2,-1,2)$ the length is $\|\mathbf{v}\| = \sqrt{4+1+4}=3$, so the unit vector in its direction is $\hat{\mathbf{v}} = \left(\frac{2}{3}, -\frac{1}{3}, \frac{2}{3}\right)$. As a check, $\|\hat{\mathbf{v}}\| = \frac{1}{3}\sqrt{4+1+4}=1$.

1.5 Beyond three dimensions: $\mathbb{R}^n$ and a data-science note

Nothing in the component definitions used the number three. A list of n real numbers

$$\mathbf{v}=(v_1, v_2, \ldots, v_n) \in \mathbb{R}^n, \qquad \|\mathbf{v}\| = \sqrt{v_1^2+\cdots+v_n^2},$$

is a vector in *n-space*, added, scaled and measured by the same formulas as before. We cannot *draw* $\mathbb{R}^n$ once n exceeds three, but the algebra is unchanged, and that is exactly what data science needs.

Remark (Data points are vectors). A record with n measured features—the pixels of an image, the word counts of a document, the readings of a sensor array—is a point of $\mathbb{R}^n$, hence a vector. Their *directions* often matter more than their lengths (a document and its doubled copy should count as identical in subject), which is why the angle between two such vectors, introduced through the dot product in Section 2, becomes the standard *cosine-similarity* measure. The geometry of this lecture is therefore not confined to two and three dimensions.

1.6 Linear combinations, independence and basis (intuition)

Adding scaled vectors, $a\mathbf{u} + b\mathbf{v} + c\mathbf{w}$, gives a *linear combination* of them. Two questions recur throughout linear algebra: is a vector *redundant*—a combination of the others—and can a given set *reach every vector*?

Note (Independence and basis, informally). A set of vectors is *linearly independent* if none of them is a linear combination of the rest—there is no redundancy. (Equivalently, and in the form you will meet again in Linear Algebra: the only linear combination that equals $\mathbf{0}$ is the one with all coefficients zero.) A *basis* is an independent set large enough to build every vector in the space, each in exactly one way. In $\mathbb{R}^3$ the standard $\mathbf{i}, \mathbf{j}, \mathbf{k}$ form the basic example: two vectors are independent unless they are parallel, and three are independent unless they lie in a common plane. The systematic *test* for independence—and the procedure that finds the coefficients of a combination—is the solution of a linear system by *Gaussian elimination*, which we develop in Lecture 5. For now the geometric picture suffices.

2 Products of vectors

There is no single "multiplication" of vectors. Two genuinely different products are useful: one returns a *scalar* and measures alignment; the other (in three dimensions) returns a *vector* perpendicular to its factors.

2.1 The scalar (dot) product

Definition 2.1 (Dot product). The *dot product* of $\mathbf{u}$ and $\mathbf{v}$ is the scalar

$$\mathbf{u} \cdot \mathbf{v} = u_1v_1 + u_2v_2 + u_3v_3 \qquad (\text{and } u_1v_1 + \cdots + u_nv_n \text{ in } \mathbb{R}^n).$$

The same number has a purely geometric description, and the link between the two is the whole reason the dot product is useful.

Theorem 2.1 (Geometric form of the dot product). If $\theta \in [0, \pi]$ is the angle between nonzero $\mathbf{u}$ and $\mathbf{v}$, then

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \, \|\mathbf{v}\| \cos\theta.$$

Proof. Place $\mathbf{u}$ and $\mathbf{v}$ tail to tail; the third side of the triangle is $\mathbf{u} - \mathbf{v}$. The law of cosines gives $\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\,\|\mathbf{u}\|\,\|\mathbf{v}\|\cos\theta$. But expanding in components, $\|\mathbf{u} - \mathbf{v}\|^2 = \sum_i (u_i - v_i)^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\sum_i u_iv_i$. Comparing the two expressions leaves $\sum_i u_iv_i = \|\mathbf{u}\|\,\|\mathbf{v}\|\cos\theta$. ∎

The dot product is commutative, distributes over addition, pulls scalars out of either factor, and a vector dotted with itself returns its length squared:

$$\mathbf{u}\cdot\mathbf{v}=\mathbf{v}\cdot\mathbf{u},\qquad \mathbf{u}\cdot(\mathbf{v}+\mathbf{w})=\mathbf{u}\cdot\mathbf{v}+\mathbf{u}\cdot\mathbf{w},\qquad (k\mathbf{u})\cdot\mathbf{v}=k(\mathbf{u}\cdot\mathbf{v}),\qquad \mathbf{v}\cdot\mathbf{v}=\|\mathbf{v}\|^2.$$

The angle between two vectors

Solving the geometric form for the cosine gives the formula that drives everything else.

Note (Angle and its sign).

$$\cos\theta=\frac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{u}\|\ \|\mathbf{v}\|}.$$

Here $\mathbf{u}$ and $\mathbf{v}$ are both nonzero, so the angle θ is defined. The sign of $\mathbf{u}\cdot\mathbf{v}$ alone reports the angle's type: positive means *acute*, negative *obtuse*, and zero a *right angle*.

Example 2.1 *(Angle between two vectors).* Find the angle between $\mathbf{u}=(1,0,1)$ and $\mathbf{v}=(1,1,0)$. The ingredients are

$$\mathbf{u}\cdot\mathbf{v}=1\cdot1+0\cdot1+1\cdot0=1,\qquad \|\mathbf{u}\|=\sqrt{2},\qquad \|\mathbf{v}\|=\sqrt{2},$$

so

$$\cos\theta=\frac{1}{\sqrt{2}\,\sqrt{2}}=\frac{1}{2}\quad\Longrightarrow\quad\theta=\frac{\pi}{3}=60^\circ.$$

The same three steps—dot the vectors, divide by the lengths, take the inverse cosine—give the angle in any dimension, and in $\mathbb{R}^n$ this cosine is the *cosine similarity* of Section 1.

Orthogonality and projection

Definition 2.2 (Orthogonality test). Vectors $\mathbf{u}$ and $\mathbf{v}$ are *orthogonal* (perpendicular) precisely when

$$\mathbf{u}\cdot\mathbf{v}=0.$$

The zero vector is taken to be orthogonal to everything.

The dot product also resolves one vector along another. The *projection* of $\mathbf{u}$ onto $\mathbf{v}$ is the shadow $\mathbf{u}$ casts on the line through $\mathbf{v}$ (Figure 3); its signed length is the *scalar projection* $\|\mathbf{u}\|\cos\theta=\dfrac{\mathbf{u}\cdot\mathbf{v}}{\|\mathbf{v}\|}$.

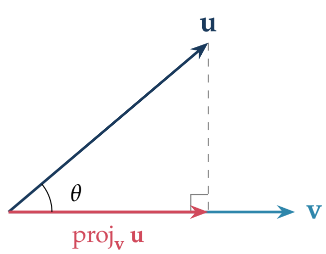

Figure 3: The vector projection of $\mathbf{u}$ onto $\mathbf{v}$.

Definition 2.3 (Vector projection). The (vector) projection of $\mathbf{u}$ onto a nonzero $\mathbf{v}$ is

$$\operatorname{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2} \mathbf{v}.$$

Example 2.2 ***(A projection).*** Project $\mathbf{u} = (4,3)$ onto $\mathbf{v} = (1,2)$. Here $\mathbf{u} \cdot \mathbf{v} = 4 + 6 = 10$ and $\|\mathbf{v}\|^2 = 1 + 4 = 5$, so

$$\operatorname{proj}_{\mathbf{v}} \mathbf{u} = \frac{10}{5}(1,2) = (2,4).$$

Remark (Physics: work). When a constant force $\mathbf{F}$ moves an object through a displacement $\mathbf{d}$, only the component of the force *along* the motion does work, and the work done is exactly the dot product $W = \mathbf{F} \cdot \mathbf{d} = \|\mathbf{F}\| \, \|\mathbf{d}\| \cos\theta$. A force perpendicular to the motion ($\theta = 90°$) does no work—the orthogonality test in physical dress.

2.2 The vector (cross) product

In three dimensions there is a second product, returning a *vector* perpendicular to both factors. It exists only in $\mathbb{R}^3$.

Definition 2.4 (Cross product). For $\mathbf{u} = (u_1, u_2, u_3)$ and $\mathbf{v} = (v_1, v_2, v_3)$ in $\mathbb{R}^3$,

$$\mathbf{u} \times \mathbf{v} = \big(u_2 v_3 - u_3 v_2,\ \ u_3 v_1 - u_1 v_3,\ \ u_1 v_2 - u_2 v_1\big).$$

The three components are awkward to memorise directly, but they are exactly the pattern of a 3×3 *determinant* with the basis vectors in the top row.

Note (The determinant mnemonic).

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = \mathbf{i} \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} - \mathbf{j} \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} + \mathbf{k} \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix},$$

where a 2×2 determinant is $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$. We use this only as a memory aid here; the determinant is defined in its own right in Lecture 5.

Theorem 2.2 (Properties of the cross product). For $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^3$ and a scalar k:

1. $\mathbf{u} \times \mathbf{v}$ is orthogonal to both $\mathbf{u}$ and $\mathbf{v}$: $\quad \mathbf{u} \cdot (\mathbf{u} \times \mathbf{v}) = 0$ and $\mathbf{v} \cdot (\mathbf{u} \times \mathbf{v}) = 0$.
2. It is *anticommutative*: $\mathbf{v} \times \mathbf{u} = -(\mathbf{u} \times \mathbf{v})$; in particular $\mathbf{u} \times \mathbf{u} = \mathbf{0}$.
3. $\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \, \|\mathbf{v}\| \sin\theta$, the area of the parallelogram with sides $\mathbf{u}$ and $\mathbf{v}$.
4. $\mathbf{u} \times \mathbf{v} = \mathbf{0}$ exactly when $\mathbf{u}$ and $\mathbf{v}$ are parallel (with the zero vector taken to be parallel to everything, covering the case where a factor is $\mathbf{0}$ and θ is undefined).
5. It distributes over addition, $\mathbf{u} \times (\mathbf{v} + \mathbf{w}) = \mathbf{u} \times \mathbf{v} + \mathbf{u} \times \mathbf{w}$, and $(k\mathbf{u}) \times \mathbf{v} = k(\mathbf{u} \times \mathbf{v})$.

The two perpendicular choices are distinguished by the *right-hand rule*: with the fingers curling from $\mathbf{u}$ to $\mathbf{v}$, the thumb points along $\mathbf{u} \times \mathbf{v}$ (Figure 4). Because the magnitude is $\|\mathbf{u}\| \, \|\mathbf{v}\| \sin\theta$, the cross product measures *turning* and *area* just as the dot product measures *alignment*.

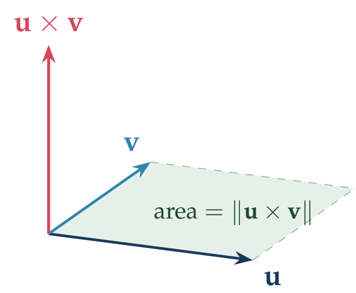

Figure 4: $\mathbf{u} \times \mathbf{v}$ is perpendicular to the plane of $\mathbf{u}, \mathbf{v}$; its length is the parallelogram's area. (Schematic: $\mathbf{u} \times \mathbf{v}$ points out of the plane of $\mathbf{u}, \mathbf{v}$, not within the page.)

Example 2.3 ***(A cross product, with checks).*** Let $\mathbf{u} = (2, 1, -1)$ and $\mathbf{v} = (1, 3, 2)$. Then

$$\mathbf{u} \times \mathbf{v} = \big(1 \cdot 2 - (-1) \cdot 3,\ \ (-1) \cdot 1 - 2 \cdot 2,\ \ 2 \cdot 3 - 1 \cdot 1\big) = (5,\ -5,\ 5).$$

The orthogonality property checks out: $\mathbf{u} \cdot (\mathbf{u} \times \mathbf{v}) = 10 - 5 - 5 = 0$ and $\mathbf{v} \cdot (\mathbf{u} \times \mathbf{v}) = 5 - 15 + 10 = 0$. Its length is $\|\mathbf{u} \times \mathbf{v}\| = \sqrt{25 + 25 + 25} = 5\sqrt{3}$, so the triangle with sides $\mathbf{u}$ and $\mathbf{v}$ has area $\frac{1}{2} \cdot 5\sqrt{3} = \frac{5\sqrt{3}}{2}$.

Remark (Physics: torque and rotation). The cross product is the natural language of rotation. The *torque* of a force $\mathbf{F}$ applied at position $\mathbf{r}$ from a pivot is $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$, and the magnetic part of the *Lorentz force* on a charge q moving with velocity $\mathbf{v}$ in a field $\mathbf{B}$ is $q\,\mathbf{v} \times \mathbf{B}$. In each case the output is perpendicular to the inputs, with a magnitude that vanishes when they are parallel—the geometry of property 3 made physical.

Optional: the scalar triple product

Remark (Volume and coplanarity (enrichment)). Combining the two products, the *scalar triple product* $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$ is a single number, equal to the 3×3 determinant whose rows are the three vectors:

$$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}.$$

Its absolute value is the *volume* of the parallelepiped built on $\mathbf{u}, \mathbf{v}, \mathbf{w}$, so the triple product is *zero* exactly when the three vectors are *coplanar*—a clean test for the three-vector independence flagged in Section 1. This is also the determinant of Lecture 5 making an early appearance: a zero determinant will mean precisely a flattened, non-invertible transformation.

Example 2.4 ***(A volume, and a coplanarity test).*** The parallelepiped built on $\mathbf{u} = (1, 2, 0)$, $\mathbf{v} = (0, 3, 1)$ and $\mathbf{w} = (2, 0, 1)$ has volume $|\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})|$. First

$$\mathbf{v} \times \mathbf{w} = \big(3 \cdot 1 - 1 \cdot 0,\ \ 1 \cdot 2 - 0 \cdot 1,\ \ 0 \cdot 0 - 3 \cdot 2\big) = (3, 2, -6),$$

and then $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = 1 \cdot 3 + 2 \cdot 2 + 0 \cdot (-6) = 7$, so the volume is 7. Equivalently, expanding the 3×3 determinant whose rows are the three vectors along its first row,

$$\begin{vmatrix} 1 & 2 & 0 \\ 0 & 3 & 1 \\ 2 & 0 & 1 \end{vmatrix} = 1 \begin{vmatrix} 3 & 1 \\ 0 & 1 \end{vmatrix} - 2 \begin{vmatrix} 0 & 1 \\ 2 & 1 \end{vmatrix} + 0 \begin{vmatrix} 0 & 3 \\ 2 & 0 \end{vmatrix} = 1(3) - 2(-2) + 0 = 7,$$

the same value by the determinant form promised above. For the coplanarity test, replace $\mathbf{u}$ by $\mathbf{w}' = \mathbf{v} + \mathbf{w} = (2, 3, 2)$, which lies in the plane of $\mathbf{v}$ and $\mathbf{w}$ by construction. As expected its triple product vanishes,

$$\mathbf{w}' \cdot (\mathbf{v} \times \mathbf{w}) = 2 \cdot 3 + 3 \cdot 2 + 2 \cdot (-6) = 0,$$

the zero signalling a collapsed, zero-volume box—exactly the three-vector dependence flagged in Section 1.

3 Lines and planes

A point fixes *where*; a vector fixes *which way*. Together they pin down the two basic flat objects of space—the line and the plane—and the products of Section 2 turn those descriptions into equations.

3.1 The equation of a line

A line is determined by one point on it and a direction to travel along. Starting from the position vector $\mathbf{r}_0$ of a known point and stepping any multiple of a direction vector $\mathbf{d}$ sweeps out the whole line (Figure 5).

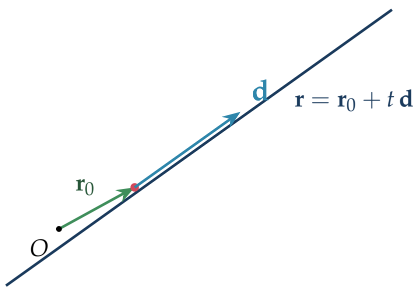

Figure 5: A line as a point $\mathbf{r}_0$ plus all multiples of a direction $\mathbf{d}$.

Definition 3.1 (Vector and parametric equations of a line). The line through the point with position vector $\mathbf{r}_0$ in the direction of a nonzero $\mathbf{d} = (d_1, d_2, d_3)$ is

$$\mathbf{r} = \mathbf{r}_0 + t\,\mathbf{d}, \qquad t \in \mathbb{R} \qquad \text{(vector form)},$$

which in coordinates is the *parametric form*

$$x = x_0 + t\,d_1, \qquad y = y_0 + t\,d_2, \qquad z = z_0 + t\,d_3.$$

Eliminating t (when no d_i is zero) gives the *symmetric form*

$$\frac{x - x_0}{d_1} = \frac{y - y_0}{d_2} = \frac{z - z_0}{d_3}.$$

In the plane $\mathbb{R}^2$ the same equation, with two components, is just the familiar straight line: writing t in terms of x recovers $y = mx + c$ with slope $m = d_2/d_1$.

Example 3.1 (*Line through two points*). The line through $A(1, 2, -1)$ and $B(3, -1, 2)$ has direction $\mathbf{d} = \overrightarrow{AB} = (2, -3, 3)$. Taking A as the base point,

$$\mathbf{r} = (1, 2, -1) + t\,(2, -3, 3), \qquad \text{i.e.} \quad x = 1 + 2t, \ \ y = 2 - 3t, \ \ z = -1 + 3t,$$

and in symmetric form $\dfrac{x-1}{2} = \dfrac{y-2}{-3} = \dfrac{z+1}{3}$. Putting $t = 1$ returns the point B, as it should.

3.2 The equation of a plane

A line needed a direction to follow; a plane is pinned down instead by the one direction it must *avoid*—a *normal* vector $\mathbf{n}$ perpendicular to every line in the plane (Figure 6). A point $\mathbf{r}$ lies in the plane through $\mathbf{r}_0$ exactly when the displacement $\mathbf{r} - \mathbf{r}_0$ lies flat, i.e. is orthogonal to $\mathbf{n}$—and orthogonality is a dot product set to zero.

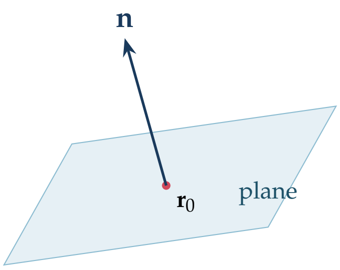

Figure 6: A plane is fixed by a point $\mathbf{r}_0$ and a normal $\mathbf{n}$.

Definition 3.2 (Equations of a plane in $\mathbb{R}^3$). The plane through the point with position vector $\mathbf{r}_0 = (x_0, y_0, z_0)$ with nonzero normal $\mathbf{n} = (a, b, c)$ is, in *normal (point-normal) form*,

$$\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0,$$

which multiplies out to the *scalar (Cartesian) form*

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0, \qquad \text{i.e.} \quad ax + by + cz = d,$$

with $d = \mathbf{n} \cdot \mathbf{r}_0$. Alternatively, given two non-parallel vectors $\mathbf{u}, \mathbf{v}$ lying in the plane, the *parametric form* is

$$\mathbf{r} = \mathbf{r}_0 + s\,\mathbf{u} + t\,\mathbf{v}, \qquad s, t \in \mathbb{R}.$$

The coefficients in the Cartesian equation are exactly the components of a normal: the plane $ax + by + cz = d$ has normal $\mathbf{n} = (a, b, c)$, read off by inspection.

Example 3.2 ***(The plane through three points).*** Find the plane through $P(1, 0, 2)$, $Q(2, 1, 0)$ and $R(0, 2, 1)$.

Two vectors in the plane. Using P as the base point,

$$\overrightarrow{PQ} = (1, 1, -2), \qquad \overrightarrow{PR} = (-1, 2, -1).$$

A normal, by the cross product. A vector perpendicular to both lies normal to the plane:

$$\mathbf{n} = \overrightarrow{PQ} \times \overrightarrow{PR} = \big(1 \cdot (-1) - (-2) \cdot 2,\ (-2) \cdot (-1) - 1 \cdot (-1),\ 1 \cdot 2 - 1 \cdot (-1)\big) = (3, 3, 3).$$

Any scalar multiple serves, so take the tidier $\mathbf{n} = (1, 1, 1)$.

The equation. Point-normal form with the point P gives

$$1(x-1) + 1(y-0) + 1(z-2) = 0 \quad\Longrightarrow\quad x + y + z = 3.$$

As a check, each of P, Q, R satisfies it: $1 + 0 + 2 = 2 + 1 + 0 = 0 + 2 + 1 = 3$. This single example uses every tool of the lecture—subtraction of points, the cross product for a normal, and the dot product hidden in the point-normal form.

3.3 Where a line meets a plane; the angle between two planes

With a line and a plane both in hand, two natural questions are pure dot-product geometry: where a line crosses a plane, and at what angle two planes meet. Each reuses the very objects built above.

Example 3.3 ***(A line meeting a plane).*** Where does the line $\mathbf{r} = (1, 2, -1) + t\,(2, -3, 3)$ of the earlier example meet the plane $x + y + z = 3$? A point of the line has coordinates $x = 1 + 2t$, $y = 2 - 3t$, $z = -1 + 3t$; substituting these into the plane equation,

$$(1 + 2t) + (2 - 3t) + (-1 + 3t) = 3 \quad\Longrightarrow\quad 2 + 2t = 3 \quad\Longrightarrow\quad t = \tfrac{1}{2}.$$

The single parameter value locates the crossing point: putting $t = \frac{1}{2}$ back into the line gives $(2, \frac{1}{2}, \frac{1}{2})$, and as a check $2 + \frac{1}{2} + \frac{1}{2} = 3$.

Note (When does a line cross a plane?). The crossing above existed because the line's direction $\mathbf{d} = (2, -3, 3)$ was *not* orthogonal to the plane's normal $\mathbf{n} = (1, 1, 1)$: here $\mathbf{d} \cdot \mathbf{n} = 2 - 3 + 3 = 2 \neq 0$, so the substitution left a genuine equation for t with one root. When instead $\mathbf{d} \cdot \mathbf{n} = 0$ the direction lies flat in the plane, and the line is either *parallel* to it (the equation for t has no solution) or *contained* in it (every t works)—the same dot product deciding which, before any arithmetic.

Example 3.4 ***(The angle between two planes).*** The angle between two planes is the angle between their normals: two planes are tilted apart exactly as much as the directions they forbid. For the plane $x + y + z = 3$, with normal $\mathbf{n}_1 = (1, 1, 1)$, and the plane $x - y + 2z = 0$, with normal $\mathbf{n}_2 = (1, -1, 2)$,

$$\cos\theta = \frac{\mathbf{n}_1 \cdot \mathbf{n}_2}{\|\mathbf{n}_1\| \, \|\mathbf{n}_2\|} = \frac{1 - 1 + 2}{\sqrt{3}\,\sqrt{6}} = \frac{2}{\sqrt{18}} = \frac{\sqrt{2}}{3} \approx 0.471,$$

so $\theta \approx 61.9^\circ$. The same dot product that measured the angle between two vectors in Section 2 now measures the angle between two flat sheets—a direct *use* of the plane equation, not just its derivation.

Note (Acute by convention; parallel planes). Two distinct planes either are *parallel*—when their normals are parallel, as for $x + y + z = 3$ and $2x + 2y + 2z = 1$—or else meet in a *line*, the case just computed. The formula returns the angle between the *chosen* normals, and reversing one normal flips the sign of the cosine; one therefore takes $|\cos\theta|$ to report the *acute* angle between the planes. The line in which two non-parallel planes meet is found by solving their two equations together—a 2×2 system, and the first hint of the simultaneous equations of Lecture 5.

Example 3.5 ***(Distance from a point to a plane).*** How far is the origin from the plane $x + y + z = 3$? Step from O along the normal direction $\mathbf{n} = (1, 1, 1)$: the point $t\,\mathbf{n} = (t, t, t)$ reaches the plane when $3t = 3$, i.e. at $t = 1$, so the foot of the perpendicular is $(1, 1, 1)$ and the distance is $\|(1, 1, 1)\| = \sqrt{3}$. The same number drops straight out of the Cartesian equation: for a plane $ax + by + cz = d$ the distance from a point $\mathbf{p} = (p_1, p_2, p_3)$ is

$$\frac{|ap_1 + bp_2 + cp_3 - d|}{\|\mathbf{n}\|} = \frac{|0 + 0 + 0 - 3|}{\sqrt{3}} = \sqrt{3},$$

the numerator measuring how far $\mathbf{p}$ sits off the plane and $\|\mathbf{n}\|$ rescaling the normal to unit length—once more the dot product doing the geometry.

Remark (Forward to Lecture 5). A line in $\mathbb{R}^2$ or a plane in $\mathbb{R}^3$ is the solution set of a *linear equation*; asking where two planes meet, or whether three planes share a point, is asking to solve a *system* of them. Whether such a system has one solution, none, or infinitely many is the geometry of intersecting, parallel and coincident planes—and the algebra that decides it, Gaussian elimination, is the heart of the next lecture.

Summary

Concept	Key idea
Vector	Tuple $\mathbf{v} = (v_1, \dots, v_n) \in \mathbb{R}^n$, closed under componentwise $+$ and scalar $\times$; pictured as a free arrow (magnitude and direction).
Operations	Componentwise $\mathbf{u} \pm \mathbf{v}$ and $k\mathbf{v}$; triangle/parallelogram rule; $k\mathbf{v}$ scales length by $\lvert k \rvert$.
Magnitude	$\lVert \mathbf{v} \rVert = \sqrt{v_1^2 + \cdots + v_n^2}$; unit vector $\hat{\mathbf{v}} = \mathbf{v} / \lVert \mathbf{v} \rVert$ (normalising).
Dot product	$\mathbf{u} \cdot \mathbf{v} = \sum u_i v_i = \lVert \mathbf{u} \rVert \lVert \mathbf{v} \rVert \cos\theta$; a scalar measuring alignment.
Angle	$\cos\theta = \dfrac{\mathbf{u} \cdot \mathbf{v}}{\lVert \mathbf{u} \rVert \lVert \mathbf{v} \rVert}$ (cosine similarity in $\mathbb{R}^n$); sign gives acute/right/obtuse.
Orthogonality	$\mathbf{u} \perp \mathbf{v} \iff \mathbf{u} \cdot \mathbf{v} = 0$.
Projection	$\operatorname{proj}_{\mathbf{v}} \mathbf{u} = \dfrac{\mathbf{u} \cdot \mathbf{v}}{\lVert \mathbf{v} \rVert^2} \mathbf{v}$.
Cross product	$\mathbf{u} \times \mathbf{v}$ (in $\mathbb{R}^3$): perpendicular to both, anticommutative, $\lVert \mathbf{u} \times \mathbf{v} \rVert = \lVert \mathbf{u} \rVert \lVert \mathbf{v} \rVert \sin\theta$ = parallelogram area; $\mathbf{0}$ iff parallel. Determinant mnemonic with $\mathbf{i}, \mathbf{j}, \mathbf{k}$ on top.
Triple product	$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$ = a 3×3 determinant; $\lvert \cdot \rvert$ is parallelepiped volume; zero iff coplanar.
Line	$\mathbf{r} = \mathbf{r}_0 + t\mathbf{d}$ (point + direction); parametric and symmetric forms.
Plane	$\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0$, i.e. $ax + by + cz = d$ with normal $\mathbf{n} = (a, b, c)$; or $\mathbf{r} = \mathbf{r}_0 + s\mathbf{u} + t\mathbf{v}$. Through three points: $\mathbf{n} = \overrightarrow{PQ} \times \overrightarrow{PR}$.

Next lecture: matrices—the arrays that store and transform vectors. We build matrix algebra and multiplication, the determinant (promised here) and the inverse, and then use Gaussian elimination to solve the linear systems whose geometry closed this lecture.

Companion material: a separate *Exercise Sheet 4* (with solutions) develops the practice problems for this unit.